Preparation of Hot Melt Extruded Supersaturating Solid Dispersions Containing Glyceryl Behenate: Effects on Processability, Supersaturation and Rate of Release

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Introduction

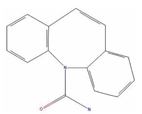
- Fusion processing of BCS Class II drugs and polymers into amorphous solid dispersion systems can result in compositions exhibiting improved dissolution and bioavailability via supersaturation of the drug following dissolution[1-3].
- These systems often include plasticizers or surfactants to improve processability and/or improve the duration of drug supersaturation following dissolution[4].
- For BCS Class II drugs with short half lives, it may be beneficial to prepare sustained release compositions provided that supersaturation can be stabilized during the duration of release.
- High molecular weight concentration enhancing polymers are desirable to produce systems that have glass transition temperatures greater than 90˚C for shelf life stability[5]. However, high molecular weight polymers require higher temperatures when melt extruded.

Study Hypothesis

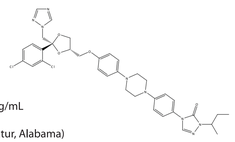
- Glyceryl behenate (Compritol® 888 ATO), commonly used in sustained release matrix tablets, may be an effective release modifier and processing aid for amorphous systems containing high molecular weight polyvinylpyrrolidone.

Materials

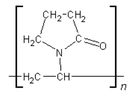
Carbamazepine (CBZ)
- Anticonvulsant
- BCS Class II Compound
- log P = 2.3
- Water Solubility = 17.7 mg/L
- Melting Point = 192˚C
- Purchased from Letco Medical (Decatur, Alabama)



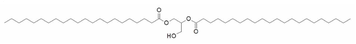
Itraconazole (ITZ)
- Triazole antifungal
- BCS Class II Compound
- pKa = 3.7
- log P = 6.9
- Melting Point = 170˚C
- pH Dependent Solubility
- Acidic Conditions = 4 µg/mL to 12 µg/mL
- Neutral Conditions = <5 ng/mL
- Purchased from Letco Medical (Decatur, Alabama)



Polyvinylpyrollidone (PVP)
- Kollidon® 30
- pH independent water soluble
- Tg = 149˚C
- Donated by BASF (Tarrytown, NY)



Glyceryl behenate
- Compritol® 888 ATO
- Mixture of behenates of glycerol
- 18% monobehenate
- 52% dibehenate
- 28% tribehenate
- Tm 72˚C
- Donated by Gattefosse (Saint Priest, France)



Methods

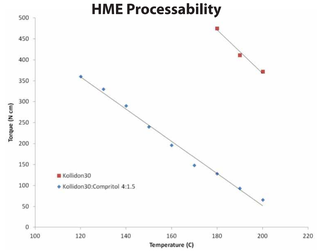
Processing
- Hot Melt Extrusion: Haake Minilab (ThermoFisher)
  - Drug, polymer and adjuvents blended and introduced to screws
  - Rotating screws force powder into heated zones and out shaped die
  - Extruded material may be milled for further processing or shaped for delivery
  - Critical process parameters
    - Screw speed – 300 RPM
    - Processing temperature
      - >180˚C without glyceryl behenate
      - 170˚C with glyceryl behenate
  - Processing torque values were collected while recirculating 10 g batches.

- Hammer Milling: Laboratory L1A Fitzmill (Fitzpatrick)
  - Configuration: 0.020 inch screen with knives forward
  - Rotational Speed: 9,000 rpm
  - Powder passed through a 60-mesh screen for analysis or further processing
- Compression: Stokes B2 Press
  - A single station was tooled and the press manually rotated
  - 5 mm (75 to 100 mg), 8 mm (200 to 300 mg) and 10mm (600mg) round tablets

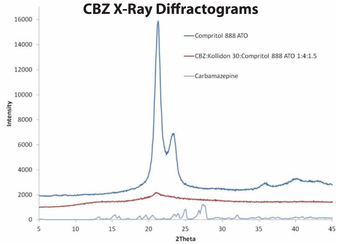
Analytical Characterization
- Modulated Differential Scanning Calorimetry
  - System: TA Instruments Model 2920
  - Sample Size: ~8 mg
  - Temperature Range: 50˚C to 200˚C
  - Ramp Rate: 10˚C/min
  - Modulation: 0.5˚C Amplitude, 40 s Period
- Raman Spectroscopy
  - System: Renshaw inVia Raman Microscope
  - Laser: 514 nm, 50 mW
  - Sampling Time: 10 seconds
  - Scan Range: 100 to 2000 cm-1
- Supersaturated Dissolution Testing
  - System: VK 7000 Dissolution Tester with VK 8000 Autosampler
  - Equivalent Amount of 100 mg API (~6X solubility CBZ, ~20X solubility ITZ)
  - Test Conditions: USP II, Method A, 50 rpm
  - Dissolution Media: 1000 mL of water (CBZ) or 0.1N HCl (ITZ)
  - pH Change Dissolution Media: 750 mL of 0.1N HCl with addition of 250 mL 0.2M Na3PO4 at 2 hours to reach pH 6.8
  - Sample Volume: 5 mL, n=3
- High Performance Liquid Chromatography (HPLC)
  - Detector: 996 Photo Diode Array Extracting at 263 nm
  - Column: Phenomenex® Luna 5 m C-18(2) 100A, 150 mm
  - CBZ Mobile Phase: 60:40 acetonitrile:water
  - ITZ Mobile Phase: 70:30:0.05 acetonitrile:water:diethanolamine
- Powder X-Ray Diffraction
  - Philips Model 1710 X-ray Diffractometer
  - Scan Range: 5˚ to 45˚
  - Step Size: 0.05˚
  - Dwell Time: 3 s
- Thermogravimetric Analysis
  - Perkin Elmer TGA 7
  - Sample Size: ~10 mg
  - Temperature Range: 50˚C to 350˚C

Results

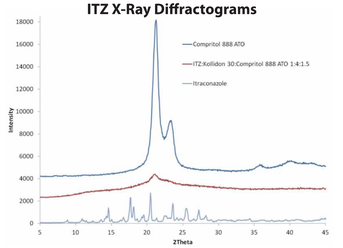
HME Processability



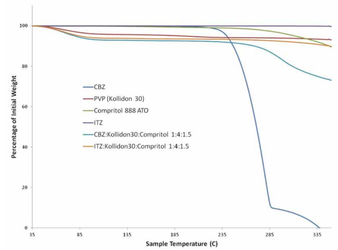
CBZ X-Ray Diffractograms



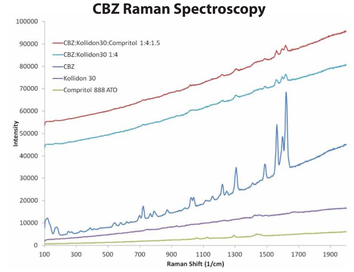
ITZ X-Ray Diffractograms



Thermogravimetric Analysis



CBZ Raman Spectroscopy



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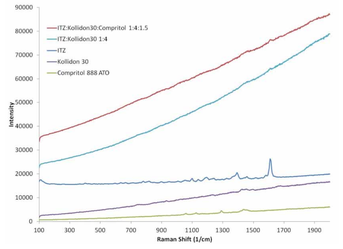

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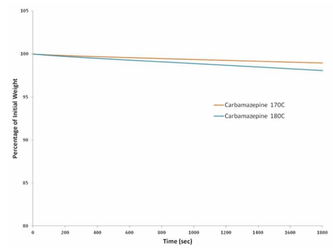
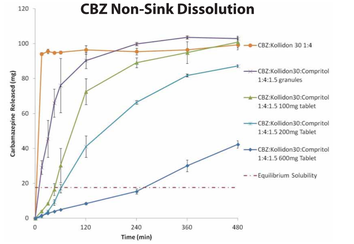
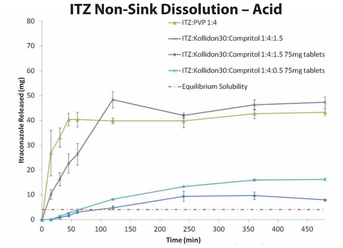


CBZ Non-Sink Dissolution



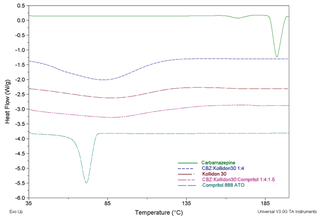
Carbamazepine dissolution rates measured by USP Method II (paddle) at 50 rpm in 1000 mL purified water maintained at 37°C (n=3).

ITZ Non-Sink Dissolution – Acid

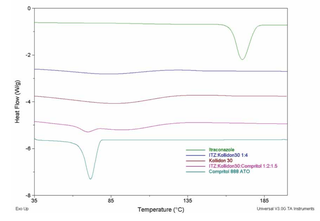


Itraconazole dissolution rates measured by USP Method II (paddle) at 50 rpm in 1000 mL 0.1N acid maintained at 37°C (n=3).

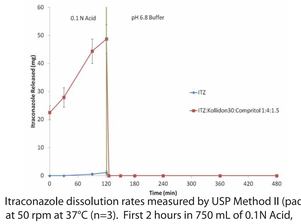
CBZ DSC Thermograms



ITZ DSC Thermograms



ITZ Non-Sink pH Change Dissolution



Itraconazole dissolution rates measured by USP Method II (paddle) at 50 rpm at 37°C (n=3). First 2 hours in 750 mL of 0.1N Acid, followed by 6 hours in 1000 mL of pH 6.8 buffer.

Discussion

- Solid dispersions containing drug, Kollidon® 30 and Compritol® 888 ATO were prepared and found to contain both itraconazole and carbamazepine in an amorphous state when analyzed by DSC, pXRD and Raman spectroscopy.
- Carbamazepine was determined to be thermally sensitive at temperatures above the Tg of Kollidon® 30. A noticeable reduction in degradation rate could be achieved by decreasing the temperature by as little as 10˚C.
- Extrusion torque values decreased substantially for Kollidon® 30 with the inclusion of Compritol® 888 ATO. This also correlated with visual observations that Kollidon® 30 solid dispersions flowed very slowly from the extruder at 180˚C, eventually overloading the extruder and resulted in low yields. Solid dispersions including Compritol® 888 ATO flowed freely from the extruder at a reduced temperature of 170˚C.
- Kollidon® 30 binary dispersions with carbamazepine and itraconazole supersaturated rapidly and were stable in non-sink dissolution conditions for 8 hours.
- When Compritol® 888 ATO was included in dispersions with carbamazepine, the rate of supersaturation was decreased, but the level of supersaturation was maintained. Tablets produced from these solid dispersions exhibited further reduction in release rate while maintaining complete supersaturation.
- Granular Compritol® 888 ATO, Kollidon® 30 and itraconazole solid dispersions exhibited similar supersaturation potential as the binary Kollidon® 30/itraconazole dispersion. However, tablets produced from these solid dispersions exhibited reduced supersaturation levels. Additionally, itraconazole precipitated from solution for solid dispersion granules following pH change.

Conclusions

- Carbamazepine and polyvinylpyrrolidone achieve stable supersaturated solutions such that the dissolution rate of carbamazepine can be controlled with the addition of Compritol. Matrices formed from this system could be used to tailor a release profile without affecting the final level of supersaturation.
- Itraconazole and polyvinylpyrrolidone achieve stable supersaturated solutions in acidic media. Reduction of the release rate with Compritol also results in a reduction in supersaturation. This system was not effective at stabilizing supersaturation following pH change.
- Compritol improves the processability of PVP dispersions as evidenced by the reduced extrusion torque and improved yield when processed with the Haake Minilab extruder.

References

1. Leuner C, Dressman J. Improving drug solubility for oral delivery using solid dispersions. Eur J Pharm Biopharm. 2000;50:47-60.
2. Miller DA, DiNunzio JC, Yang W, McGinity JW, Williams RO. Enhanced in vivo absorption of itraconazole via stabilization of supersaturation following acidic-to-neutral pH transition. Drug Dev Ind Pharm. 2008;34:890-902.
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Acknowledgements

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